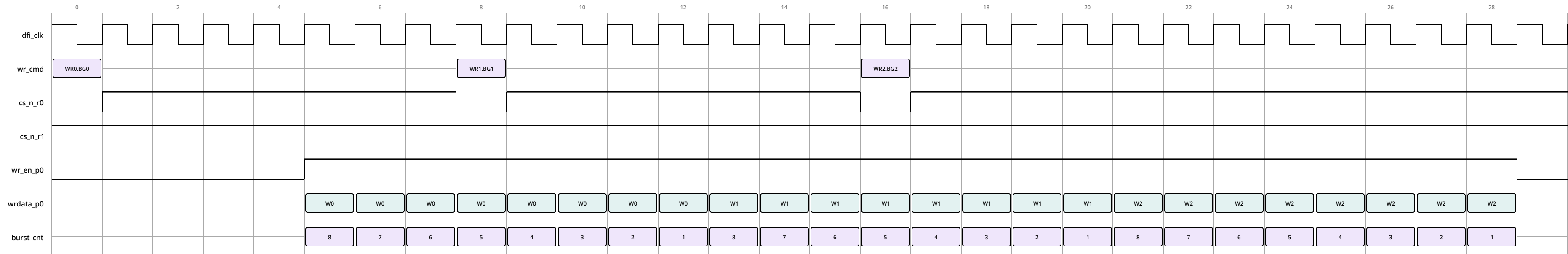


Case 1 - 3 continuous WR, SAME rank, diff-BG (8 CK apart)

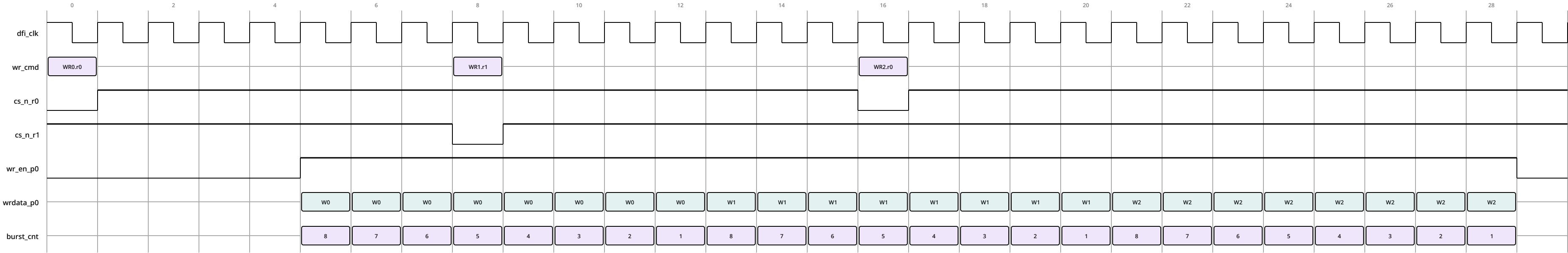
1:1 gear; WR0/1/2 @ 0/8/16; WL=tphy_wrlat+tphy_wrdata=5 (const shift); each burst=8 beats(64B)



spacing = tCCD_S = dqFree = 8. same-BG would be tCCD_L_WR=48. WL just shifts the whole stream; WR1/WR2 latency pipelines run DURING WR0's data -> back-to-back, no re-count.

Case 2 - 3 WR CROSS-rank (r0, r1, r0), 8 CK apart

1:1 gear; WR0(r0)@0, WR1(r1)@8, WR2(r0)@16; tphy_wrlat/wrdata SAME for all ranks -> same WL=5



cmd handoff tRTRS=2 but dqFree=8 still gates -> data 8 apart, no faster (DQ-bound). win = dodges same-BG tCCD_L_WR=48. CS toggles rank; SAME tphy -> bursts tile identically.